Lesson 58: Policy Gradient Synthesis

AI/ML Mathematics

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1 Motivation and Overview

Consider deploying a policy gradient algorithm on a complex continuous control task – a humanoid robot learning to walk. The practitioner faces a cascade of design choices: Should the gradient use the full return G_t, the TD error δ_t, or the GAE estimator? Should the update follow the vanilla gradient, the natural gradient, or a trust-region step? What convergence rate can we expect, and will the algorithm find a global or merely a local optimum? Each of Lessons 54–57 addressed one piece of this puzzle in isolation. The purpose of this capstone lesson is to unify these results into a single, coherent framework, to compare the methods rigorously, and to chart the path from the theoretical foundations of Phase 02 to the deep RL algorithms of Phase 04.

We present the *unified policy gradient equation*, which expresses every algorithm from Lessons 54–57 as a single expression $\nabla_\theta J(\theta) = \mathbb{E}_\pi[\sum_t \nabla_\theta \log \pi_\theta(A_t \mid S_t)\, \Psi_t]$ under different choices of the signal Ψ_t. We then develop the *variance reduction taxonomy*, proving which choices are unbiased and quantifying their variance. A convergence landscape section collects the convergence rates and optimality conditions. Finally, two preview sections connect the natural gradient to trust-region methods (TRPO, PPO) and the policy gradient to entropy-regularized RL (SAC), showing how Phase 02 foundations lead directly into Phase 04.

RL/ML Connection

- **TRPO** [Schulman et al., 2015]: Constrains the policy update to a KL-divergence trust region, building on the natural gradient and the performance difference lemma derived here.
- **PPO** [Schulman et al., 2017]: Approximates TRPO with a clipped surrogate objective; its theoretical grounding traces back to the unified advantage formulation.
- **SAC** [Haarnoja et al., 2018]: Uses the soft policy gradient theorem derived from the entropy-regularized objective previewed here.
- **Natural actor-critic** [Peters and Schaal, 2008]: Combines the natural gradient with compatible features; the unified framework shows this is the Fisher-preconditioned advantage estimator.

Prerequisites. This lesson synthesizes all of Phase 02. The reader must be familiar with: the policy gradient theorem and baseline invariance (Lesson 54, Theorem 5.1 and Proposition 7.1), the REINFORCE estimator and optimal baseline (Lesson 55, Theorems 3.2 and 5.4), the Fisher information matrix and natural gradient (Lesson 56, Definitions 4.1 and 5.1, Theorem 5.2), the actor-critic architecture and GAE (Lesson 57, Definitions 3.1 and 6.1, Theorem 6.2). From the MDP lessons, we assume knowledge of Bellman equations, value and policy iteration (Lessons 39–46). From TD learning, we assume familiarity with TD(0), TD(λ), and function approximation (Lessons 47–53).

Learning objectives.

- **State** the unified policy gradient equation and identify the signal Ψ_t corresponding to each algorithm (REINFORCE, baseline, actor-critic, GAE, natural gradient).
- **Prove** that all valid signal choices yield unbiased gradients or characterize the bias for biased estimators.
- **Compare** the variance and convergence rates of different policy gradient methods.

- **Derive** the Kakade-Langford performance difference lemma and the conservative policy iteration bound.
- **Explain** how the natural gradient leads to trust-region methods (TRPO, PPO) and how entropy regularization leads to SAC.
- **Construct** a dependency map of Phase 02 showing how MDP, TD, and policy gradient results interconnect.

2 References

- Sutton and Barto [2018, Chapter 13] – Policy gradient methods; unified treatment of REINFORCE, actor-critic, and baselines.
- Kakade and Langford [2002] – Conservative policy iteration and the performance difference lemma connecting policy gradient to trust regions.
- Schulman et al. [2016] – GAE derivation and bias-variance analysis for advantage estimators.
- Schulman et al. [2015] – TRPO: trust-region optimization using KL divergence constraints derived from the natural gradient.
- Haarnoja et al. [2018] – SAC: maximum entropy RL with the soft policy gradient theorem.
- Agarwal et al. [2021, Chapters 5–8] – Modern convergence theory for policy gradient methods including global optimality results.

3 The Unified Policy Gradient Framework

Every policy gradient algorithm studied in Lessons 54–57 can be cast as an instance of a single template. This section makes the unification precise.

Definition 3.1 (General Policy Gradient Signal)**.** Let $\tau = (S_0, A_0, R_1, S_1, A_1, R_2, \ldots)$ be a trajectory generated by π_θ. A *policy gradient signal* is a random variable Ψ_t (possibly depending on the entire trajectory) such that the gradient estimator

$$\hat{g}(\tau) = \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(A_t \mid S_t)\, \Psi_t$$

satisfies $\mathbb{E}[\hat{g}(\tau)] = c \cdot \nabla_\theta J(\theta)$ for some constant $c > 0$ independent of θ. When $c = 1$ and the equality holds exactly, Ψ_t is called an *unbiased* signal.

Definition 3.2 (Signal Choices)**.** The following are the principal signal choices studied across Lessons 54–57:

(i) **Full return (REINFORCE, Lesson 55):** $\Psi_t = G_t = \sum_{k=0}^{T-t-1} \gamma^k R_{t+k+1}$.

(ii) **Return with baseline (Lesson 55):** $\Psi_t = G_t - b(S_t)$ for any function $b : \mathcal{S} \to \mathbb{R}$.

(iii) **Action-value function:** $\Psi_t = Q^\pi(S_t, A_t)$.

(iv) **Advantage function (Lesson 55):** $\Psi_t = A^\pi(S_t, A_t) = Q^\pi(S_t, A_t) - V^\pi(S_t)$.

(v) **TD error (Lesson 57):** $\Psi_t = \delta_t^\pi = R_{t+1} + \gamma\, V^\pi(S_{t+1}) - V^\pi(S_t)$.

(vi) **GAE (Lesson 57):** $\Psi_t = \hat{A}_t^{\mathrm{GAE}(\gamma,\lambda)} = \sum_{k=0}^{T-t-1} (\gamma\lambda)^k\, \delta_{t+k}^{\hat{V}}$.

Unified Policy Gradient Equation

All policy gradient algorithms share the form

$$\nabla_\theta J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{\infty} \gamma^t \, \nabla_\theta \log \pi_\theta(A_t \mid S_t) \, \Psi_t\right],$$

where the choice of Ψ_t determines the specific algorithm:

Signal Ψ_t	**Algorithm**	**Lesson**
G_t	REINFORCE	55
$G_t - b(S_t)$	REINFORCE with baseline	55
$Q^\pi(S_t, A_t)$	Policy Gradient Theorem	54
$A^\pi(S_t, A_t)$	Advantage PG	55
δ_t^π	One-step actor-critic	57
$\hat{A}_t^{\mathrm{GAE}(\gamma,\lambda)}$	GAE actor-critic	57

All signals except $\delta_t^{\hat{V}}$ with $\hat{V} \neq V^\pi$ and GAE with $\lambda < 1$ and $\hat{V} \neq V^\pi$ yield unbiased gradient estimators.

Theorem 3.3 (Unbiasedness of Signal Choices)**.** *Let Ψ_t be any of the signals (i)–(v) in Definition 3.2 (with the true value function V^π used in (v)). Then*

$$\mathbb{E}_{\pi_\theta}[\nabla_\theta \log \pi_\theta(A_t \mid S_t) \, \Psi_t \mid S_t] = \sum_a \nabla_\theta \pi_\theta(a \mid S_t) \, Q^\pi(S_t, a).$$

In particular, all five signals yield the same expected gradient.

Proof. We treat each case.

Case (i): $\Psi_t = G_t$. By the tower property, conditioning on (S_t, A_t): $\mathbb{E}[G_t \mid S_t, A_t] = Q^\pi(S_t, A_t)$ (Lesson 55, Lemma 3.2). Then

$$\begin{aligned}
\mathbb{E}[\nabla_\theta \log \pi_\theta(A_t \mid S_t) \, G_t \mid S_t] &= \mathbb{E}[\nabla_\theta \log \pi_\theta(A_t \mid S_t) \, Q^\pi(S_t, A_t) \mid S_t] \\
&= \sum_a \pi_\theta(a \mid S_t) \, \nabla_\theta \log \pi_\theta(a \mid S_t) \, Q^\pi(S_t, a) \\
&= \sum_a \nabla_\theta \pi_\theta(a \mid S_t) \, Q^\pi(S_t, a).
\end{aligned}$$

Case (ii): $\Psi_t = G_t - b(S_t)$. The $b(S_t)$ term contributes $b(S_t) \sum_a \nabla_\theta \pi_\theta(a \mid S_t) = b(S_t) \cdot 0 = 0$ by the score function zero-mean property (Lesson 54, Lemma 4.2). The result follows from case (i).

Case (iii): $\Psi_t = Q^\pi(S_t, A_t)$. This is a special case of (i) where G_t is replaced by its conditional expectation. The calculation is identical to the last two lines of case (i).

Case (iv): $\Psi_t = A^\pi(S_t, A_t)$. Since $A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s)$ and $V^\pi(s)$ is a baseline, this follows from cases (iii) and (ii) combined.

Case (v): $\Psi_t = \delta_t^\pi$. By Theorem 4.2 of Lesson 57, $\mathbb{E}[\delta_t^\pi \mid S_t, A_t] = A^\pi(S_t, A_t)$. Therefore:

$$\mathbb{E}[\nabla_\theta \log \pi_\theta(A_t \mid S_t) \, \delta_t^\pi \mid S_t] = \mathbb{E}[\nabla_\theta \log \pi_\theta(A_t \mid S_t) \, A^\pi(S_t, A_t) \mid S_t],$$

which equals $\sum_a \nabla_\theta \pi_\theta(a \mid S_t) \, Q^\pi(S_t, a)$ by case (iv). □

Proposition 3.4 (Bias of Approximate Signals). *When the TD error uses an approximate value function $\hat{V}_w \neq V^\pi$, the signal $\Psi_t = \delta_t^{\hat{V}_w}$ introduces a bias. Specifically, the bias of the per-step gradient estimator at state s is*

$$\text{bias}(s) = \sum_a \nabla_\theta \pi_\theta(a \mid s) \left[\gamma \sum_{s'} \mathcal{P}(s' \mid s, a) \left(\hat{V}_w(s') - V^\pi(s')\right) - \left(\hat{V}_w(s) - V^\pi(s)\right)\right].$$

The baseline component $\hat{V}_w(s) - V^\pi(s)$ vanishes by the score zero-mean property, leaving only the bootstrapping error:

$$\text{bias}(s) = \gamma \sum_a \nabla_\theta \pi_\theta(a \mid s) \sum_{s'} \mathcal{P}(s' \mid s, a) \left(\hat{V}_w(s') - V^\pi(s')\right).$$

Proof. From Proposition 4.4 of Lesson 57, $\mathbb{E}[\delta_t^{\hat{V}_w} \mid S_t = s, A_t = a] = A^\pi(s,a) + \gamma \sum_{s'} \mathcal{P}(s' \mid s,a)(\hat{V}_w(s') - V^\pi(s')) - (\hat{V}_w(s) - V^\pi(s))$. Computing $\mathbb{E}[\nabla_\theta \log \pi_\theta(A_t \mid s) \cdot \delta_t^{\hat{V}_w} \mid S_t = s]$ and using case (iv) of Theorem 3.3 for the A^π term gives the first expression. For the state-dependent error term $\hat{V}_w(s) - V^\pi(s)$:

$$\sum_a \nabla_\theta \pi_\theta(a \mid s) \left[-(\hat{V}_w(s) - V^\pi(s))\right] = -(\hat{V}_w(s) - V^\pi(s)) \cdot \nabla_\theta \underbrace{\sum_a \pi_\theta(a \mid s)}_{=1} = 0,$$

leaving only the successor-state error term. □

RL/ML Connection

Proposition 3.4 reveals a fundamental asymmetry: the state-level error $\hat{V}_w(s) - V^\pi(s)$ is harmless (it acts as a baseline), but the successor-state error $\hat{V}_w(s') - V^\pi(s')$ introduces genuine bias through bootstrapping. This is why REINFORCE with an imperfect baseline remains unbiased, while actor-critic methods with an imperfect critic are biased. The bias is the price paid for reduced variance through bootstrapping.

4 Variance Reduction Taxonomy

We now collect and compare all variance reduction techniques studied in Lessons 54–57, establishing a taxonomy that clarifies the design space for policy gradient algorithms.

Definition 4.1 (Variance of a Policy Gradient Estimator). For a signal Ψ_t and a single trajectory τ, the total variance of the gradient estimator $\hat{g}(\tau) = \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(A_t \mid S_t)\, \Psi_t$ is

$$\text{Var}(\hat{g}) = \mathbb{E}\left[\|\hat{g} - \mathbb{E}[\hat{g}]\|^2\right].$$

We decompose the total variance into contributions from each time step and cross-time correlations.

Theorem 4.2 (Variance Ordering of Signal Choices). *Under the assumption that $\hat{V} = V^\pi$ (so all estimators are unbiased), the following variance ordering holds for a single time step at state s:*

$$\text{Var}(\textit{TD error}) \leq \text{Var}(\textit{advantage}) \leq \text{Var}(\textit{return with baseline}) \leq \text{Var}(\textit{return}).$$

More precisely, for any fixed state s:

(i) $\text{Var}(G_t - V^\pi(S_t)) \leq \text{Var}(G_t)$ *(baseline reduces variance).*

(ii) $\mathrm{Var}(A^\pi(S_t, A_t)) \le \mathrm{Var}(G_t - V^\pi(S_t))$ *(replacing the random G_t with the deterministic Q^π eliminates future trajectory randomness).*

(iii) $\mathrm{Var}(\delta_t^\pi) \le \mathrm{Var}(A^\pi(S_t, A_t))$ *(the TD error introduces randomness only from the next transition, not from the action-value function).*

Proof. (i) By the variance decomposition $\mathrm{Var}(G_t) = \mathrm{Var}(G_t - V^\pi(S_t)) + 2\,\mathrm{Cov}(G_t - V^\pi(S_t), V^\pi(S_t)) + \mathrm{Var}(V^\pi(S_t))$. Since $V^\pi(S_t)$ is determined by S_t: $\mathrm{Var}(G_t) = \mathrm{Var}(G_t - c) + \mathrm{Var}(c) + 2\,\mathrm{Cov}(G_t - c, c) \ge \mathrm{Var}(G_t - c)$ when $c = V^\pi(S_t)$ is a constant given S_t (the inequality follows from the standard result that subtracting the conditional mean minimizes conditional variance). Conditioning on S_t, since $\mathbb{E}[G_t \mid S_t] \ne V^\pi(S_t)$ in general (it depends on the action distribution), we use the simpler argument: $\mathrm{Var}(\hat{g}_b(s)) = \mathrm{Var}(\hat{g}(s)) - b^2\,\mathrm{Var}(\text{score})$ terms from Lesson 55, Lemma 5.1, with the specific analysis depending on b. The key inequality is that any baseline can only reduce variance compared to no baseline, since the optimal baseline achieves the minimum.

(ii) Conditioning on (S_t, A_t): $\mathrm{Var}(G_t - V^\pi(S_t) \mid S_t, A_t) = \mathrm{Var}(G_t \mid S_t, A_t)$, which captures the randomness of future transitions. Since $A^\pi(S_t, A_t) = Q^\pi(S_t, A_t) - V^\pi(S_t)$ is a deterministic function of (S_t, A_t), $\mathrm{Var}(A^\pi(S_t, A_t) \mid S_t, A_t) = 0$. By the law of total variance: $\mathrm{Var}(G_t - V^\pi(S_t) \mid S_t) = \mathbb{E}[\mathrm{Var}(G_t \mid S_t, A_t) \mid S_t] + \mathrm{Var}(Q^\pi(S_t, A_t) - V^\pi(S_t) \mid S_t)$, so $\mathrm{Var}(G_t - V^\pi(S_t) \mid S_t) \ge \mathrm{Var}(A^\pi(S_t, A_t) \mid S_t)$.

(iii) The advantage $A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s)$ is a deterministic function of (s, a). The TD error $\delta_t^\pi = r(S_t, A_t) + \gamma V^\pi(S_{t+1}) - V^\pi(S_t)$ is random (through S_{t+1}) with $\mathbb{E}[\delta_t^\pi \mid S_t, A_t] = A^\pi(S_t, A_t)$. Conditioning on S_t, the variance comparison depends on the specific MDP structure. For the per-step gradient estimator, the variance of $\nabla \log \pi \cdot \delta_t^\pi$ conditional on S_t involves both the action randomness and the next-state randomness, while the variance of $\nabla \log \pi \cdot A^\pi$ involves only action randomness. The TD error adds next-state variance but the single-step nature bounds the total variance below that of multi-step returns. □

Remark 4.3. The ordering in Theorem 4.2 applies to the variance of the *signal* Ψ_t, not directly to the variance of the full gradient estimator (which also depends on cross-time correlations). In practice, the variance of the full trajectory-based estimator is dominated by the per-step signal variance, so the ordering is a good guide for algorithm design.

Variance Reduction Taxonomy

The variance reduction techniques in policy gradient methods form four categories, each reducing a different source of randomness:

Technique	Mechanism	Bias	Lesson Reference
Baseline $b(s)$	Subtract state-dependent constant	None	Lesson 55, Thm. 5.4
Actor-critic ($\hat{V}_w$)	Replace return with bootstrapped estimate	$O(\lVert\epsilon\rVert)$	Lesson 57, Sec. 4
GAE(γ, λ)	Exponential weighting of multi-step TD	$O(\lVert\epsilon\rVert/(1 - \gamma\lambda))$	Lesson 57, Thm. 7.1
Natural gradient (F^{-1})	Fisher preconditioning of parameter update	None	Lesson 56, Thm. 5.2

The first three reduce variance in the *signal* Ψ_t; the fourth reduces variance in the *parameter update direction* by aligning it with the geometry of the policy manifold.

Example 4.4 (Combining All Variance Reduction Techniques)**.** A modern policy gradient implementation typically combines all four techniques. Given a batch of trajectories under π_θ:

1. Fit a critic $\hat{V}_w$ to minimize $\sum_t (G_t - \hat{V}_w(S_t))^2$.
2. Compute GAE advantages: $\hat{A}_t^{\text{GAE}} = \sum_{k=0}^{T-t-1} (\gamma\lambda)^k \delta_{t+k}^{\hat{V}_w}$.
3. Estimate the gradient: $\hat{g} = \frac{1}{N} \sum_{n=1}^{N} \sum_t \nabla_\theta \log \pi_\theta(A_t^n \mid S_t^n)\, \hat{A}_t^{n,\text{GAE}}$.
4. Compute the natural gradient direction: $\tilde{g} = F(\theta)^{-1}\, \hat{g}$.
5. Update: $\theta \leftarrow \theta + \alpha\, \tilde{g}$.

This combines baseline subtraction (through $\hat{V}_w$ in the advantage), bootstrapping (through the TD errors in GAE), the λ tradeoff (through GAE), and Fisher preconditioning (through the natural gradient step).

5 Convergence Landscape

This section collects the convergence results from Lessons 55–57, places them in a unified framework, and addresses the question of local versus global optimality.

Definition 5.1 (Convergence Rate Classes)**.** For a sequence of iterates $\{\theta_k\}_{k=0}^K$, we measure convergence by the quantity $\min_{0\le k\le K} \mathbb{E}[\|\nabla_\theta J(\theta_k)\|^2]$:

- $O(1/\sqrt{K})$ **rate:** The minimum expected squared gradient norm decreases as $1/\sqrt{K}$. This is the standard rate for stochastic gradient methods with bounded variance.
- $O(1/K)$ **rate:** Faster convergence, achievable with variance reduction or natural gradient preconditioning.

Theorem 5.2 (Convergence Rate of Vanilla Policy Gradient)**.** *Under the regularity conditions of Lesson 55 (Assumption 7.1): Lipschitz gradient with constant L, bounded variance $\mathbb{E}[\|\hat{g} - \nabla J\|^2] \le \sigma^2$, and step size $\alpha_k = \alpha/\sqrt{K}$ with $\alpha = \Theta(1/(L + \sigma/\sqrt{K}))$, the vanilla policy gradient satisfies*

$$\min_{0\le k\le K} \mathbb{E}\big[\|\nabla_\theta J(\theta_k)\|^2\big] = O\left(\frac{L\cdot(J^* - J(\theta_0)) + \sigma^2}{\sqrt{K}}\right),$$

where $J^* = \sup_\theta J(\theta)$.

Proof. By L-smoothness: $J(\theta_{k+1}) \ge J(\theta_k) + \nabla J(\theta_k)^\top(\theta_{k+1} - \theta_k) - \frac{L}{2}\|\theta_{k+1} - \theta_k\|^2$. Substituting $\theta_{k+1} - \theta_k = \alpha_k \hat{g}_k$ and taking expectations:

$$\mathbb{E}[J(\theta_{k+1})] \ge \mathbb{E}[J(\theta_k)] + \alpha_k \mathbb{E}[\|\nabla J(\theta_k)\|^2] - \frac{L\alpha_k^2}{2}\,(\mathbb{E}[\|\nabla J(\theta_k)\|^2] + \sigma^2).$$

Rearranging and summing from $k = 0$ to $K - 1$:

$$\sum_{k=0}^{K-1} \alpha_k (1 - \frac{L\alpha_k}{2}) \mathbb{E}[\|\nabla J(\theta_k)\|^2] \le J^* - J(\theta_0) + \frac{L\sigma^2}{2} \sum_{k=0}^{K-1} \alpha_k^2.$$

With $\alpha_k = c/\sqrt{K}$ for appropriate c: the left side is $\Theta(\sqrt{K}) \cdot \min_k \mathbb{E}[\|\nabla J\|^2]$ and the right side is $O(J^* - J_0 + \sigma^2/\sqrt{K} \cdot L)$, yielding the stated $O(1/\sqrt{K})$ rate. □

Theorem 5.3 (Convergence Rate of Natural Policy Gradient)**.** *Under the additional assumption that the Fisher information matrix satisfies $\mu_{\min} I \preceq F(\theta) \preceq \mu_{\max} I$ for all θ (bounded condition number $\kappa = \mu_{\max}/\mu_{\min}$), the natural policy gradient with exact gradients and step size $\alpha = 1/(\mu_{\max})$ satisfies*

$$J^* - J(\theta_K) = O\left(\frac{1}{K}\right).$$

Proof. The natural gradient update $\theta_{k+1} = \theta_k + \alpha F(\theta_k)^{-1}\nabla J(\theta_k)$ achieves per-step improvement:

$$J(\theta_{k+1}) - J(\theta_k) \geq \alpha \nabla J^\top F^{-1} \nabla J - \frac{L\alpha^2}{2}\|F^{-1}\nabla J\|^2.$$

Since $\|F^{-1}\nabla J\|^2 \leq \frac{1}{\mu_{\min}^2}\|\nabla J\|^2$ and $\nabla J^\top F^{-1}\nabla J \geq \frac{1}{\mu_{\max}}\|\nabla J\|^2$, choosing $\alpha = \mu_{\min}^2/(L)$ gives:

$$J(\theta_{k+1}) - J(\theta_k) \geq \frac{\mu_{\min}^2}{2L\mu_{\max}}\|\nabla J(\theta_k)\|^2.$$

If $J(\theta)$ additionally satisfies a gradient domination condition $J^* - J(\theta) \leq C\|\nabla J(\theta)\|^2$ (which holds for softmax parameterizations under certain conditions), then $J^* - J(\theta_{k+1}) \leq (1 - \mu_{\min}^2/(2LC\mu_{\max}))(J^* - J(\theta_k))$, giving linear (i.e., $O(1/K)$) convergence. □

Convergence Rate Comparison

Method	Rate (stochastic)	Key Condition
Vanilla PG	$O(1/\sqrt{K})$	L-smooth objective
Natural PG (exact)	$O(1/K)$	Bounded Fisher condition number
Natural PG (stochastic)	$O(1/\sqrt{K})$	Adds gradient noise

The natural gradient achieves a faster rate in the exact-gradient setting due to preconditioning. With stochastic gradients, the improvement manifests as better constants (depending on κ) rather than a different rate class.

Theorem 5.4 (Local vs. Global Optimality)**.** *The following characterization holds for policy gradient methods:*

(i) ***General parameterization:*** *Policy gradient methods converge to* stationary points *of $J(\theta)$. The objective $J(\theta)$ is generally non-concave, so stationary points may be local maxima or saddle points.*

(ii) ***Softmax (tabular) parameterization:*** *For finite MDPs with the direct softmax parameterization $\pi_\theta(a \mid s) \propto \exp(\theta_{s,a})$, every stationary point of $J(\theta)$ corresponds to a globally optimal policy π^*.*

Proof. (i) This follows from Theorem 7.3 of Lesson 55: under Robbins-Monro step sizes, REINFORCE converges to the set $\{\theta : \nabla_\theta J(\theta) = 0\}$.

(ii) This is the result of Mei et al. [2020]. The key insight is that for the softmax parameterization, $J(\theta)$ satisfies the *gradient domination* property: for all θ,

$$J(\pi^*) - J(\theta) \leq \frac{1}{1-\gamma}\left\|\frac{d^{\pi^*}}{d^{\pi_\theta}}\right\|_\infty \cdot \|\nabla_\theta J(\theta)\|_{F^{-1}}.$$

Here $\|\cdot\|_{F^{-1}}$ denotes the norm under the inverse Fisher metric. Any stationary point $\nabla_\theta J(\theta) = 0$ therefore satisfies $J(\pi^*) = J(\theta)$, meaning the corresponding policy is globally optimal. The proof relies on the fact that the softmax parameterization has full support ($\pi_\theta(a \mid s) > 0$ for all s, a) and that the policy can approximate any distribution arbitrarily closely as $\|\theta\| \to \infty$. □

Global Optimality Requires Full Support

The global optimality result for softmax parameterization relies critically on the full-support property. Neural network parameterizations generally do not satisfy gradient domination, and policy gradient methods can converge to suboptimal stationary points. Adding entropy regularization (Section 7) helps by keeping the policy away from determinism and improving the optimization landscape.

6 Performance Difference Lemma and Trust Regions

The natural gradient (Lesson 56) controls policy changes via a second-order KL approximation. The Kakade-Langford performance difference lemma provides an *exact* identity relating the performance of two policies, from which trust-region bounds can be derived.

Theorem 6.1 (Performance Difference Lemma)**.** *For any two policies π and π':*

$$J(\pi') - J(\pi) = \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi'}}\left[\sum_a \pi'(a \mid s)\, A^{\pi}(s,a)\right],$$

where $d^{\pi'}$ is the (normalized) discounted state visitation distribution under π'.

Proof. Define $V^{\pi'}(s) - V^{\pi}(s)$ and expand using the Bellman equation for π'.

Step 1. By the Bellman equation under π':

$$V^{\pi'}(s) = \sum_a \pi'(a \mid s)\left[r(s,a) + \gamma \sum_{s'} \mathcal{P}(s' \mid s,a)\, V^{\pi'}(s')\right].$$

Step 2. Subtract $V^{\pi}(s)$ on both sides and add/subtract $\sum_a \pi'(a \mid s)\, Q^{\pi}(s,a)$:

$$\begin{aligned} V^{\pi'}(s) - V^{\pi}(s) &= \sum_a \pi'(a \mid s)\left[Q^{\pi}(s,a) - V^{\pi}(s)\right] + \gamma \sum_{s'} \mathbb{P}^{\pi'}(s' \mid s)\left[V^{\pi'}(s') - V^{\pi}(s')\right] \\ &= \sum_a \pi'(a \mid s)\, A^{\pi}(s,a) + \gamma \sum_{s'} \mathbb{P}^{\pi'}(s' \mid s)\left[V^{\pi'}(s') - V^{\pi}(s')\right], \end{aligned}$$

where we used $Q^{\pi}(s,a) = r(s,a) + \gamma \sum_{s'} \mathcal{P}(s' \mid s,a)\, V^{\pi}(s')$ and $\mathbb{P}^{\pi'}(s' \mid s) = \sum_a \pi'(a \mid s)\, \mathcal{P}(s' \mid s,a)$.

Step 3. Unrolling the recursion (as in the proof of the policy gradient theorem, Lesson 54, Theorem 5.1):

$$V^{\pi'}(s_0) - V^{\pi}(s_0) = \sum_{t=0}^{\infty} \gamma^t \sum_s \mathbb{P}(S_t = s \mid s_0, \pi') \sum_a \pi'(a \mid s)\, A^{\pi}(s,a).$$

Step 4. Using the (normalized) discounted state visitation distribution $d^{\pi'}(s) = (1-\gamma)\sum_t \gamma^t \mathbb{P}(S_t = s \mid s_0, \pi')$:

$$J(\pi') - J(\pi) = V^{\pi'}(s_0) - V^{\pi}(s_0) = \frac{1}{1-\gamma}\sum_s d^{\pi'}(s) \sum_a \pi'(a \mid s)\, A^{\pi}(s,a). \qquad \square$$

Remark 6.2. The performance difference lemma is an *exact* identity, not an approximation. The difficulty in using it directly is that the right side involves $d^{\pi'}$, the state distribution under the *new* policy, which is unknown before executing π'. The key contribution of Kakade and Langford [2002] and Schulman et al. [2015] is to bound the difference between $d^{\pi'}$ and d^{π}, yielding a usable lower bound on $J(\pi')$ that depends only on quantities computable under π.

Definition 6.3 (Surrogate Objective)**.** The *local surrogate objective* replaces $d^{\pi'}$ with d^{π} in the performance difference lemma:

$$L_\pi(\pi') = J(\pi) + \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^\pi}\left[\sum_a \pi'(a \mid s)\, A^\pi(s,a)\right].$$

This is a first-order approximation: $L_\pi(\pi) = J(\pi)$ and $\nabla_{\theta'} L_{\pi_\theta}(\pi_{\theta'})\big|_{\theta'=\theta} = \nabla_\theta J(\theta)$.

Theorem 6.4 (Conservative Policy Improvement Bound)**.** *Let* $\epsilon = \max_s |\mathbb{E}_{a\sim\pi'(\cdot|s)}[A^\pi(s,a)]|$. *Define the maximum KL divergence* $D^{\mathrm{KL}}_{\max}(\pi,\pi') = \max_s \mathrm{KL}(\pi(\cdot \mid s)\|\pi'(\cdot \mid s))$. *Then:*

$$J(\pi') \geq L_\pi(\pi') - \frac{2\gamma\,\epsilon}{(1-\gamma)^2}\, D^{\mathrm{KL}}_{\max}(\pi,\pi').$$

Proof. The proof bounds the difference $|d^{\pi'}(s) - d^\pi(s)|$ using the data processing inequality for total variation distance.

Step 1. From the exact identity (Theorem 6.1):

$$J(\pi') - L_\pi(\pi') = \frac{1}{1-\gamma}\sum_s \big(d^{\pi'}(s) - d^\pi(s)\big)\sum_a \pi'(a \mid s)\, A^\pi(s,a).$$

Step 2. Bounding the inner sum by ϵ and the total variation by $\|d^{\pi'} - d^\pi\|_{\mathrm{TV}} \leq \frac{\gamma}{1-\gamma}\max_s \|\pi'(\cdot \mid s) - \pi(\cdot \mid s)\|_{\mathrm{TV}}$:

$$|J(\pi') - L_\pi(\pi')| \leq \frac{2\epsilon}{(1-\gamma)} \cdot \frac{\gamma}{1-\gamma}\max_s \|\pi'(\cdot \mid s) - \pi(\cdot \mid s)\|_{\mathrm{TV}}.$$

Step 3. By Pinsker's inequality, $\|\pi - \pi'\|_{\mathrm{TV}} \leq \sqrt{\frac{1}{2}\mathrm{KL}(\pi\|\pi')}$, so $\|\pi - \pi'\|_{\mathrm{TV}} \leq \sqrt{\frac{1}{2}D^{\mathrm{KL}}_{\max}}$. Squaring and applying AM-GM gives the bound with $D^{\mathrm{KL}}_{\max}$ (the tighter version uses the square root, but the linear form suffices for deriving trust-region algorithms). □

From Natural Gradient to Trust Regions

The conservative policy improvement bound shows that maximizing the surrogate $L_\pi(\pi')$ subject to a KL divergence constraint $D^{\mathrm{KL}}_{\max}(\pi,\pi') \leq \delta$ guarantees monotonic improvement:

$$J(\pi') \geq J(\pi) + [\text{improvement from } L_\pi] - \frac{2\gamma\,\epsilon}{(1-\gamma)^2}\,\delta.$$

This is exactly the TRPO optimization problem. The natural gradient (Lesson 56) solves this problem to second order: the constraint $\Delta\theta^\top F(\theta)\Delta\theta \leq \delta$ approximates $\mathbb{E}_{s\sim d^\pi}[\mathrm{KL}(\pi_\theta\|\pi_{\theta+\Delta\theta})] \leq \delta/2$.

Preview: TRPO and PPO

TRPO [Schulman et al., 2015] directly solves the constrained optimization:

$$\theta_{k+1} = \arg\max_\theta\ \mathbb{E}_{s\sim d^{\pi_{\theta_k}}}\left[\sum_a \pi_\theta(a \mid s)\,\hat{A}^{\pi_{\theta_k}}(s,a)\right] \quad \text{s.t.} \quad \mathbb{E}_{s\sim d^{\pi_{\theta_k}}}[\mathrm{KL}(\pi_{\theta_k}\|\pi_\theta)] \leq \delta,$$

using conjugate gradients with Fisher-vector products (avoiding explicit Fisher matrix construction) and a line search to satisfy the constraint.

PPO [Schulman et al., 2017] approximates the KL constraint with a clipped surrogate:

$$L^{\mathrm{CLIP}}(\theta) = \mathbb{E}\Big[\min\Big(r_t(\theta)\,\hat{A}_t,\ \mathrm{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon)\,\hat{A}_t\Big)\Big],$$

where $r_t(\theta) = \pi_\theta(A_t \mid S_t)/\pi_{\theta_{\text{old}}}(A_t \mid S_t)$ is the importance ratio. The clipping removes the incentive to move r_t far from 1, implicitly enforcing a trust region without computing KL divergences. The full derivation of TRPO and PPO appears in Phase 04 (Lessons 32–33).

7 Connection to Maximum Entropy RL

A natural extension of the policy gradient framework adds an entropy bonus to the objective, encouraging exploration and improving optimization landscapes.

Definition 7.1 (Entropy-Regularized Objective)**.** The *entropy-regularized* (maximum entropy) objective is

$$J_{\text{ent}}(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{\infty} \gamma^t \big(r(S_t, A_t) + \alpha\, \mathcal{H}(\pi_\theta(\cdot \mid S_t))\big)\right],$$

where $\mathcal{H}(\pi(\cdot \mid s)) = -\sum_a \pi(a \mid s) \log \pi(a \mid s)$ is the Shannon entropy and $\alpha > 0$ is the *temperature parameter* controlling the exploration-exploitation tradeoff.

Definition 7.2 (Soft Value Functions)**.** The *soft value function* and *soft action-value function* under π_θ are:

$$V^\pi_{\text{soft}}(s) = \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} \gamma^t \big(r(S_t, A_t) + \alpha\, \mathcal{H}(\pi(\cdot \mid S_t))\big) \,\middle|\, S_0 = s\right],$$
$$Q^\pi_{\text{soft}}(s,a) = r(s,a) + \gamma \mathbb{E}_{s' \sim \mathcal{P}(\cdot \mid s,a)}\big[V^\pi_{\text{soft}}(s')\big].$$

These satisfy the *soft Bellman equation*: $V^\pi_{\text{soft}}(s) = \mathbb{E}_{a\sim\pi}[Q^\pi_{\text{soft}}(s,a) - \alpha \log \pi(a \mid s)]$.

Theorem 7.3 (Soft Policy Gradient Theorem)**.** *The gradient of the entropy-regularized objective is*

$$\nabla_\theta J_{\text{ent}}(\theta) = \mathbb{E}_{s\sim d^{\pi_\theta},\, a\sim\pi_\theta}\big[\nabla_\theta \log \pi_\theta(a \mid s) \cdot \big(Q^\pi_{\text{soft}}(s,a) - \alpha \log \pi_\theta(a \mid s) - V^\pi_{\text{soft}}(s)\big)\big].$$

Equivalently, defining the soft advantage $A^\pi_{\text{soft}}(s,a) = Q^\pi_{\text{soft}}(s,a) - \alpha \log \pi_\theta(a \mid s) - V^\pi_{\text{soft}}(s)$*:*

$$\nabla_\theta J_{\text{ent}}(\theta) = \mathbb{E}_{s\sim d^\pi,\, a\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a \mid s)\, A^\pi_{\text{soft}}(s,a)].$$

Proof. We adapt the proof of the standard policy gradient theorem (Lesson 54, Theorem 5.1) to the entropy-augmented reward $\tilde{r}(s,a) = r(s,a) + \alpha\mathcal{H}(\pi(\cdot \mid s))$.

Step 1. The gradient of the soft value function satisfies the recursion (paralleling equation (5.2) of Lesson 54):

$$\nabla_\theta V^\pi_{\text{soft}}(s) = \sum_a \nabla_\theta \pi_\theta(a \mid s)\, Q^\pi_{\text{soft}}(s,a) + \sum_a \pi_\theta(a \mid s)\, \nabla_\theta Q^\pi_{\text{soft}}(s,a) + \alpha \nabla_\theta \mathcal{H}(\pi_\theta(\cdot \mid s)).$$

Step 2. The entropy gradient is: $\nabla_\theta \mathcal{H} = -\sum_a \nabla_\theta \pi_\theta (\log \pi_\theta + 1) = -\sum_a \nabla_\theta \pi_\theta \log \pi_\theta$, since $\sum_a \nabla_\theta \pi_\theta = 0$. Substituting and combining with the first term:

$$\sum_a \nabla_\theta \pi_\theta \cdot Q^\pi_{\text{soft}} - \alpha \sum_a \nabla_\theta \pi_\theta \log \pi_\theta = \sum_a \nabla_\theta \pi_\theta \cdot (Q^\pi_{\text{soft}} - \alpha \log \pi_\theta).$$

Step 3. The term $\nabla_\theta Q^\pi_{\text{soft}}$ introduces the recursion through $V^\pi_{\text{soft}}(s')$, just as in the standard proof. Unrolling and using the discounted state visitation distribution:

$$\nabla_\theta J_{\text{ent}} = \sum_s d^\pi(s) \sum_a \nabla_\theta \pi_\theta(a \mid s)(Q^\pi_{\text{soft}}(s,a) - \alpha \log \pi_\theta(a \mid s)).$$

Converting to log-derivative form and subtracting the baseline $V^\pi_{\text{soft}}(s)$ (which vanishes by the score zero-mean property) gives the result. □

Soft Optimal Policy

The optimal policy under the entropy-regularized objective satisfies

$$\pi^*(a \mid s) = \frac{\exp(Q^*_{\text{soft}}(s,a)/\alpha)}{\sum_{a'} \exp(Q^*_{\text{soft}}(s,a')/\alpha)},$$

which is a Boltzmann (softmax) distribution over actions with temperature α. As $\alpha \to 0$, the soft optimal policy converges to the standard (greedy) optimal policy. As $\alpha \to \infty$, the policy converges to the uniform distribution.

Preview: Soft Actor-Critic (SAC)

SAC [Haarnoja et al., 2018] implements the soft policy gradient theorem with three neural networks:

- A *soft critic* $Q_w(s,a)$ trained via soft Bellman backup: $Q_w(s,a) \to r + \gamma\, \mathbb{E}_{s'}[V_{\bar{w}}(s')]$ where $V_{\bar{w}}(s) = \mathbb{E}_{a\sim\pi_\theta}[Q_{\bar{w}}(s,a) - \alpha \log \pi_\theta(a \mid s)]$.
- An *actor* π_θ updated via the soft policy gradient with reparameterization trick for continuous actions.
- Automatic *temperature* α tuning via a dual gradient descent on the target entropy constraint.

The full derivation appears in Phase 04 (Lesson 35). The key advantage of SAC over vanilla actor-critic is improved exploration (through the entropy term) and off-policy learning (through the soft Bellman backup), which dramatically improves sample efficiency.

8 Phase 02 Grand Summary

This section provides a comprehensive summary of the entire Phase 02 arc, covering all twenty lessons across three subdomains.

Phase 02 Arc: From MDPs to Policy Gradients

Phase 02 developed reinforcement learning theory in three stages:
Stage 1 – MDP Foundations (Lessons 39–46): Defined the sequential decision-making problem and derived exact solution methods. The Bellman equations provide the recursive structure; value iteration and policy iteration provide algorithms with known convergence guarantees; the LP formulation connects MDPs to mathematical programming.
Stage 2 – TD Learning (Lessons 47–53): Developed sample-based methods that learn from experience without a model. TD(0) bootstraps from one-step transitions; TD(λ) interpolates between TD and Monte Carlo; SARSA and Q-learning provide on-policy and off-policy control; function approximation scales to large state spaces but introduces the deadly triad (function approximation + bootstrapping + off-policy).
Stage 3 – Policy Gradients (Lessons 54–58): Directly optimized the policy using gradient ascent on the expected return. The policy gradient theorem provides the foundational identity; REINFORCE and baselines address variance; the natural gradient addresses parameterization sensitivity; actor-critic and GAE trade bias for variance. This

synthesis lesson unified the framework and connected to Phase 04.

L#	Topic	Key Results
Subdomain 1: Markov Decision Processes		
39	MDP formulation	State/action spaces, transition kernel, reward, discount factor, policy, trajectory
40	Bellman equations	Bellman expectation/optimality equations, contraction property of Bellman operator
41	Value iteration	$V_{k+1} = \mathcal{T}V_k$, convergence rate $\|V_k - V^*\|_\infty \leq \gamma^k \|V_0 - V^*\|_\infty$
42	Error bounds	Approximate VI error bounds, simulation lemma, sensitivity to model perturbations
43	Average reward MDPs	Gain-bias Bellman equation, Blackwell optimality, relative value iteration
44	Policy iteration	Monotone improvement, finite convergence for finite MDPs, modified PI
45	LP formulation	Primal/dual LP for MDPs, occupancy measures, equivalence to VI/PI
46	Finite-horizon MDPs	Backward induction, time-varying policies, non-stationary Bellman equations
Subdomain 2: Temporal Difference Learning		
47	TD(0) prediction	$V(S_t) \leftarrow V(S_t) + \alpha\delta_t$, convergence under Robbins-Monro
48	n-step TD	n-step returns, bias-variance tradeoff, forward/backward views
49	TD(λ)	Eligibility traces, equivalence of forward and backward views, convergence
50	SARSA	On-policy control, ϵ-greedy exploration, convergence to optimal Q
51	Q-learning	Off-policy control, convergence with probability 1, maximization bias
52	Function approximation	Linear TD, semi-gradient methods, projected Bellman equation
53	Deadly triad	Divergence examples, Baird's counterexample, conditions for stability
Subdomain 3: Policy Gradients		
54	PG theorem	$\nabla J = \mathbb{E}_{d^\pi}[\nabla \log \pi \cdot Q^\pi]$, compatible function approx.
55	REINFORCE	Monte Carlo PG, baseline invariance, optimal baseline $b^*(s)$, convergence
56	Natural PG	Fisher information, $\tilde{\nabla} J = F^{-1}\nabla J$, covariance, compatible features
57	Actor-critic / GAE	TD advantage, n-step estimators, $\hat{A}^{\text{GAE}} = \sum(\gamma\lambda)^k \delta_{t+k}$, two-timescale
58	Synthesis	Unified framework, PDL, trust region preview, MaxEnt preview

Dependency Graph of Phase 02

The following diagram shows how the key results build on each other:

Bellman equations (L40)
$\downarrow$
Value/Policy iteration (L41, L44) $\longrightarrow$ LP formulation (L45)
$\downarrow$
TD(0) prediction (L47) $\longrightarrow$ n-step TD (L48) $\longrightarrow$ TD(λ) (L49)
$\downarrow$ $\downarrow$
SARSA / Q-learning (L50–51) Func. approx. (L52) $\rightarrow$ Deadly triad (L53)
$\downarrow$
PG Theorem (L54)
$\downarrow$
REINFORCE + baselines (L55)
$\swarrow$ $\searrow$
Natural PG (L56) Actor-critic / GAE (L57)
$\searrow$ $\swarrow$
Synthesis (L58)
$\downarrow$
Phase 04: TRPO (L32), PPO (L33), SAC (L35)

9 Exercises

Exercise 9.1 (Unified Signal Verification). ($*$)
Starting from the policy gradient theorem $\nabla_\theta J = \mathbb{E}_{d^\pi,\pi}[\nabla_\theta \log \pi_\theta(a \mid s)\, Q^\pi(s,a)]$ (Lesson 54, Theorem 5.1), verify that using $\Psi_t = \delta_t^\pi = r(S_t, A_t) + \gamma V^\pi(S_{t+1}) - V^\pi(S_t)$ yields the same expected gradient. You may use the results $\mathbb{E}[\delta_t^\pi \mid S_t, A_t] = A^\pi(S_t, A_t)$ and $\mathbb{E}_{a\sim\pi}[A^\pi(s,a)] = 0$.

Exercise 9.2 (Bias Quantification for Actor-Critic). ($*$)
Consider a two-state MDP with $\mathcal{S} = \{s_1, s_2\}$, a single action, deterministic transitions $s_1 \rightarrow s_2 \rightarrow s_1$, rewards $r(s_1) = 1$, $r(s_2) = 0$, and $\gamma = 0.9$. Suppose the critic estimates $\hat{V}(s_1) = 5.0$ and $\hat{V}(s_2) = 4.0$.

(a) Compute the true values $V^\pi(s_1)$ and $V^\pi(s_2)$.

(b) Compute $\mathbb{E}[\delta_t^{\hat{V}} \mid S_t = s_1]$ and compare to $A^\pi(s_1, a) = 0$.

(c) Verify that the bias equals $\gamma(\hat{V}(s_2) - V^\pi(s_2)) - (\hat{V}(s_1) - V^\pi(s_1))$, as predicted by Proposition 3.4.

Exercise 9.3 (Performance Difference Lemma Application). ($**$)
Consider a bandit (single state) with $\mathcal{A} = \{a_1, a_2, a_3\}$, rewards $r(a_1) = 1$, $r(a_2) = 3$, $r(a_3) = 2$, and current policy $\pi(a_1) = 0.5$, $\pi(a_2) = 0.3$, $\pi(a_3) = 0.2$.

(a) Compute V^π, $Q^\pi(a_i)$, and $A^\pi(a_i)$ for each action.

(b) A new policy π' puts $\pi'(a_1) = 0.1$, $\pi'(a_2) = 0.7$, $\pi'(a_3) = 0.2$. Use the performance difference lemma (Theorem 6.1) to compute $J(\pi') - J(\pi)$ directly.

(c) Verify your answer by computing $J(\pi')$ and $J(\pi)$ independently.

(d) Compute the KL divergence $\mathrm{KL}(\pi \| \pi')$ and the bound from Theorem 6.4. Is the bound tight?

Exercise 9.4 (Soft Policy Gradient for a Bandit). ($**$)
For a two-action bandit with $r(a_1) = 0$, $r(a_2) = 1$, temperature $\alpha = 0.5$, and softmax policy $\pi_\theta(a_2) = \sigma(\theta) = 1/(1 + e^{-\theta})$:

(a) Compute the entropy-regularized objective $J_{\text{ent}}(\theta)$ as a function of θ.

(b) Find the optimal θ^* by setting $\nabla_\theta J_{\text{ent}} = 0$.

(c) Verify that π_{θ^*} matches the Boltzmann form $\pi^*(a) \propto \exp(r(a)/\alpha)$.

(d) Compare θ^* to the unregularized optimum ($\alpha = 0$).

Exercise 9.5 (Convergence Rate Comparison)**.** $(***)$
Consider an MDP where $J(\theta)$ is L-smooth and the Fisher information matrix satisfies $\mu I \preceq F(\theta) \preceq \kappa\mu I$ for constants $\mu > 0$ and $\kappa \geq 1$.

(a) For the vanilla gradient with step size $\alpha = 1/(2L)$ and exact gradients, show the per-step improvement is at least $\frac{1}{4L}\|\nabla J\|^2$.

(b) For the natural gradient with step size $\alpha = 1/(2L)$, show the per-step improvement is at least $\frac{1}{4L\kappa\mu}\|\nabla J\|^2$.

(c) Derive conditions on κ and μ under which the natural gradient has better per-step improvement than the vanilla gradient.

Hint: Use $\nabla J^\top F^{-1} \nabla J \geq \frac{1}{\kappa\mu}\|\nabla J\|^2$ for part (b).

Exercise 9.6 (Designing a Policy Gradient Algorithm)**.** $(***)$
You are given a continuous control task with a 20-dimensional action space, stochastic transitions, and sparse rewards (nonzero only near the goal). Design a policy gradient algorithm by specifying:

(a) The signal Ψ_t (from Definition 3.2), with justification.

(b) The value of λ for GAE, with justification.

(c) Whether to use the vanilla or natural gradient, with justification.

(d) Whether to add entropy regularization, with justification.

For each choice, cite the specific theorem or result from Lessons 54–58 that supports your design decision.

10 Summary

- The **unified policy gradient equation** $\nabla J = \mathbb{E}[\sum_t \nabla \log \pi \cdot \Psi_t]$ subsumes all algorithms from Lessons 54–57 under different signal choices $\Psi_t \in \{G_t,\ G_t - b(s),\ Q^\pi,\ A^\pi,\ \delta_t,\ \hat{A}^{\text{GAE}}\}$.
- All valid signals yield **unbiased gradients** except when bootstrapping with an approximate critic ($\hat{V} \neq V^\pi$), which introduces bias through the successor-state value error. The baseline component remains harmless by the score zero-mean property.
- **Variance reduction** operates through four mechanisms: baselines (subtract $b(s)$, zero bias), bootstrapping (use TD estimates, nonzero bias), λ-weighting (GAE interpolation), and Fisher preconditioning (natural gradient in parameter space).
- Vanilla policy gradient converges at rate $O(1/\sqrt{K})$; the natural gradient achieves $O(1/K)$ with exact gradients under gradient domination. For **softmax tabular parameterizations**, all stationary points are globally optimal.

- The **performance difference lemma** $J(\pi') - J(\pi) = \frac{1}{1-\gamma}\mathbb{E}_{d^{\pi'}}[\sum_a \pi' A^\pi]$ provides an exact identity for comparing policies. Combined with KL divergence bounds, it yields the trust-region optimization framework (TRPO).
- **Entropy regularization** adds $\alpha\mathcal{H}(\pi)$ to the reward, producing the soft policy gradient theorem and the Boltzmann optimal policy $\pi^* \propto \exp(Q^*/\alpha)$. This framework underlies SAC.
- **Phase 02 arc**: MDP foundations (Bellman equations, VI, PI) → TD learning (sample-based prediction and control, function approximation) → Policy gradients (direct optimization, variance reduction, geometry). This progression moves from model-based exact methods to model-free approximate methods with increasing generality.
- The three pillars of Phase 02 – **dynamic programming**, **temporal difference learning**, and **policy gradients** – converge in Phase 04 (deep RL), where neural network function approximators enable scaling to high-dimensional problems.

11 Further Reading

- **Phase 04, Lesson 32** derives TRPO in full detail, solving the constrained optimization problem from Theorem 6.4 using conjugate gradients and line search.
- **Phase 04, Lesson 33** develops PPO's clipped surrogate objective, showing how it approximates the trust-region constraint while being simpler to implement.
- **Phase 04, Lesson 34** derives the deterministic policy gradient theorem (Silver et al., 2014), extending the stochastic policy gradient of Lesson 54 to deterministic policies via a limiting argument.
- **Phase 04, Lesson 35** develops the full SAC algorithm from the soft policy gradient theorem (Theorem 7.3), including automatic temperature tuning.
- Agarwal et al. [2021] provides modern convergence theory for policy gradient methods, including global convergence results for natural policy gradient and entropy regularization.
- Kakade and Langford [2002] and Kakade [2003] give the original conservative policy iteration analysis and sample complexity bounds.
- Mei et al. [2020] proves global convergence of softmax policy gradient with an explicit convergence rate.

References

Alekh Agarwal, Nan Jiang, Sham M. Kakade, and Wen Sun. *Reinforcement Learning: Theory and Algorithms*. RLTA Draft, 2021.

Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. *Proceedings of the 35th International Conference on Machine Learning*, pages 1861–1870, 2018.

Sham Kakade and John Langford. Approximately optimal approximate reinforcement learning. *Proceedings of the 19th International Conference on Machine Learning*, pages 267–274, 2002.

Sham Machandranath Kakade. *On the Sample Complexity of Reinforcement Learning*. PhD thesis, University College London, 2003.

Jincheng Mei, Chenjun Xiao, Csaba Szepesvári, and Dale Schuurmans. On the global convergence rates of softmax policy gradient methods. *Proceedings of the 37th International Conference on Machine Learning*, pages 6820–6829, 2020.

Jan Peters and Stefan Schaal. Natural actor-critic. *Neurocomputing*, 71(7–9):1180–1190, 2008.

John Schulman, Sergey Levine, Pieter Abbeel, Michael Jordan, and Philipp Moritz. Trust region policy optimization. *Proceedings of the 32nd International Conference on Machine Learning*, pages 1889–1897, 2015.

John Schulman, Philipp Moritz, Sergey Levine, Michael Jordan, and Pieter Abbeel. High-dimensional continuous control using generalized advantage estimation. *Proceedings of the International Conference on Learning Representations*, 2016.

John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2nd edition, 2018.